

Investment Deck Apr 2021

The Future of Farming is Indoors



By 2050, the world's population is predicted to reach

9.7B

and to feed
everyone would require
increase in food output by

70%

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Indoor Climate Controls are fragmented, manual, and labor intensive

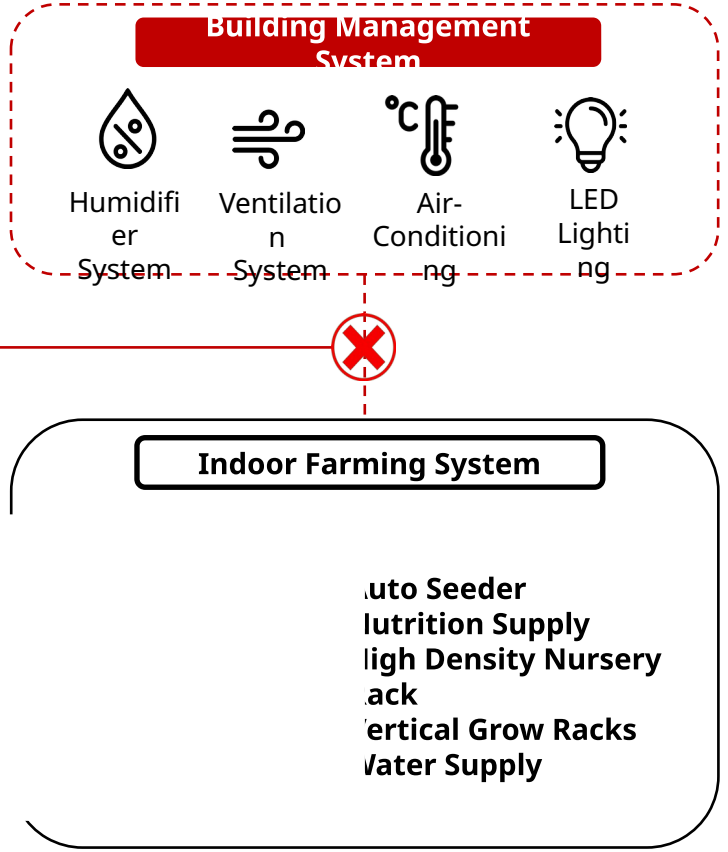
OPTIMAL CROP CONDITIONS DEPENDS ON SEVERAL FACTORS

Day-time Temp: 20-24C
Night-time Temp: 10-12C
Humidity: Low humidity
(Crown): 18C
Light Intensity: 800-1200 micro-mol/m2/s PPF



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Climate At Your Service

Lack of an end-to-end Building Management System for Indoor Farms



Inability to provide 24/7, automated, and optimized climate conditions

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Agri-tech remains key area of investment in Southeast Asia

Farm Management Software, Sensing & IOT is the top 5 most invested Agri-tech category in the region (2019)



$423M

Invested

+33%

YoY Investment Growth

+41%

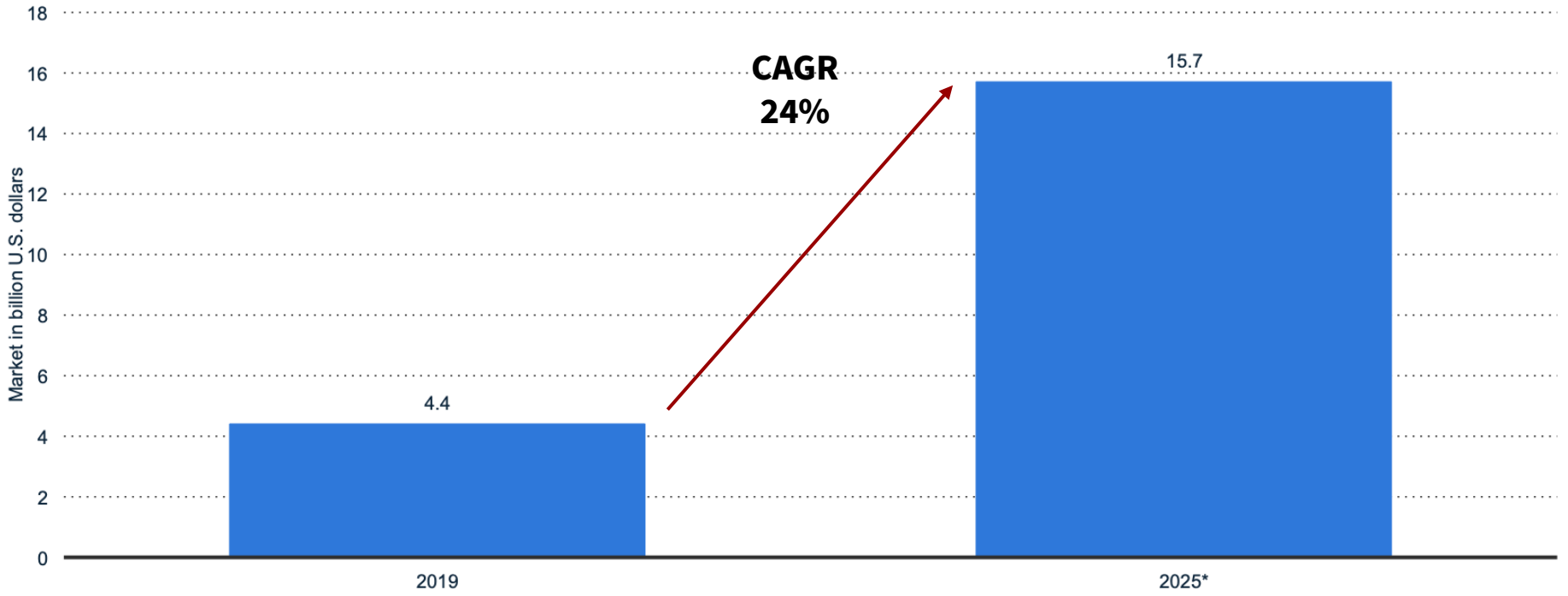
YoY Deal Growth

Source: 2020 , Agfunder

Market: Growth

Projected vertical farming market worldwide in 2019 and 2025 (in billion U.S. dollars)

Global vertical farming market projection 2019 & 2025

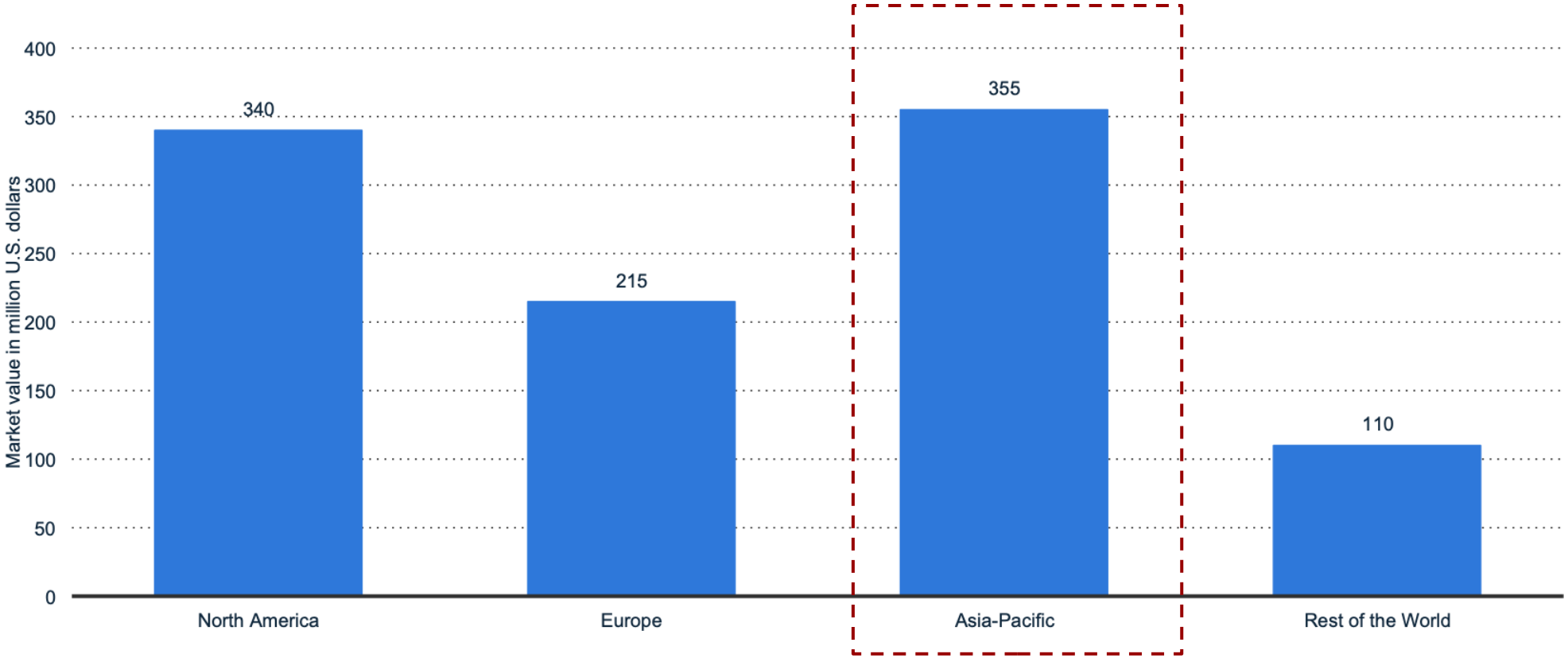


Source: Statista

SOM

Market value of vertical farming worldwide in 2015, by region (in million U.S. dollars)

Global vertical farming market value 2015, by region



Source: Statista

Ambient Platform: Climate-At-Your-Service



Connected.

Integrated with equipment and control systems

Centralized

Automated.

Machine-learning Optimization

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Video Demo:
Ambient MVP Pilot

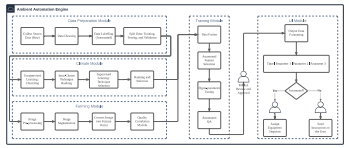


Ambient At A Glance

Connectivity

Machine Learning

Key Equipment



Automation Engine

Smart Sensors Control Units Edge Nodes

Ambient Dashboard

CO2 Regulator

Humidifier/ Dehumidifier

Smart Thermostat

Building Management System

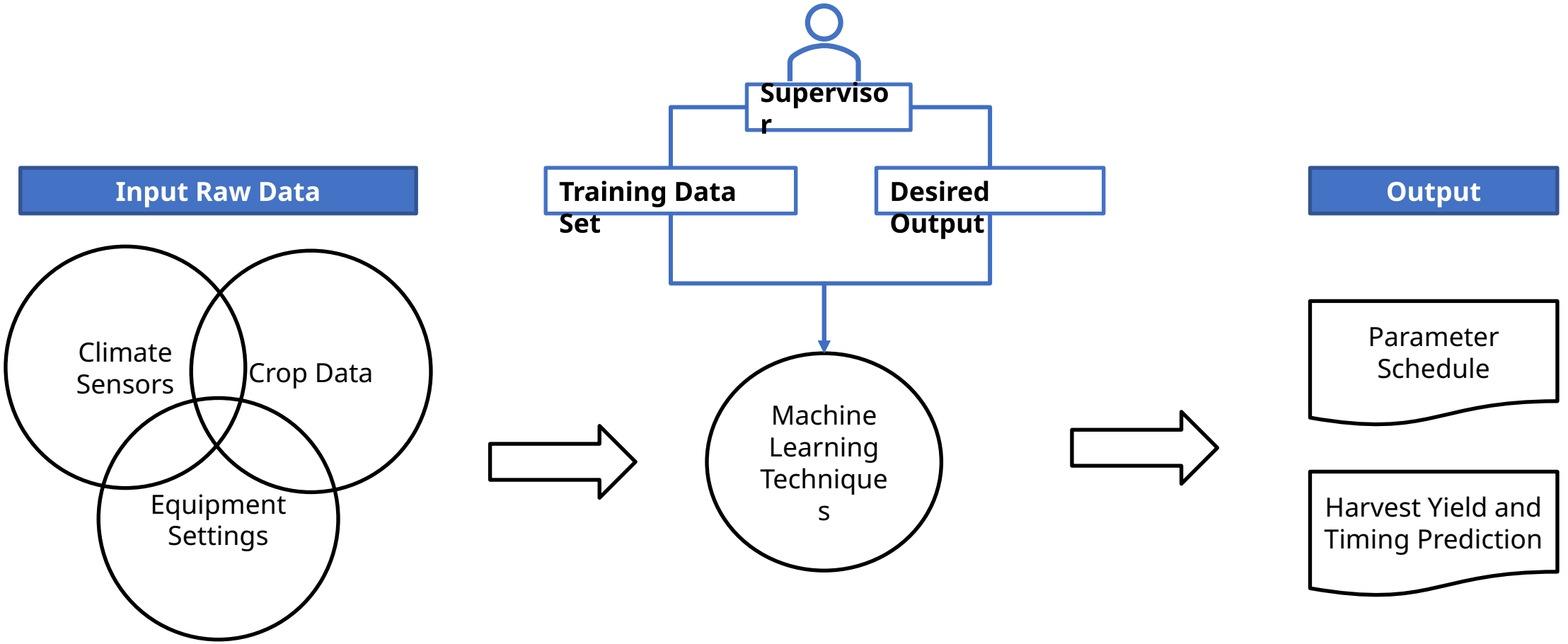
Indoor Farming System

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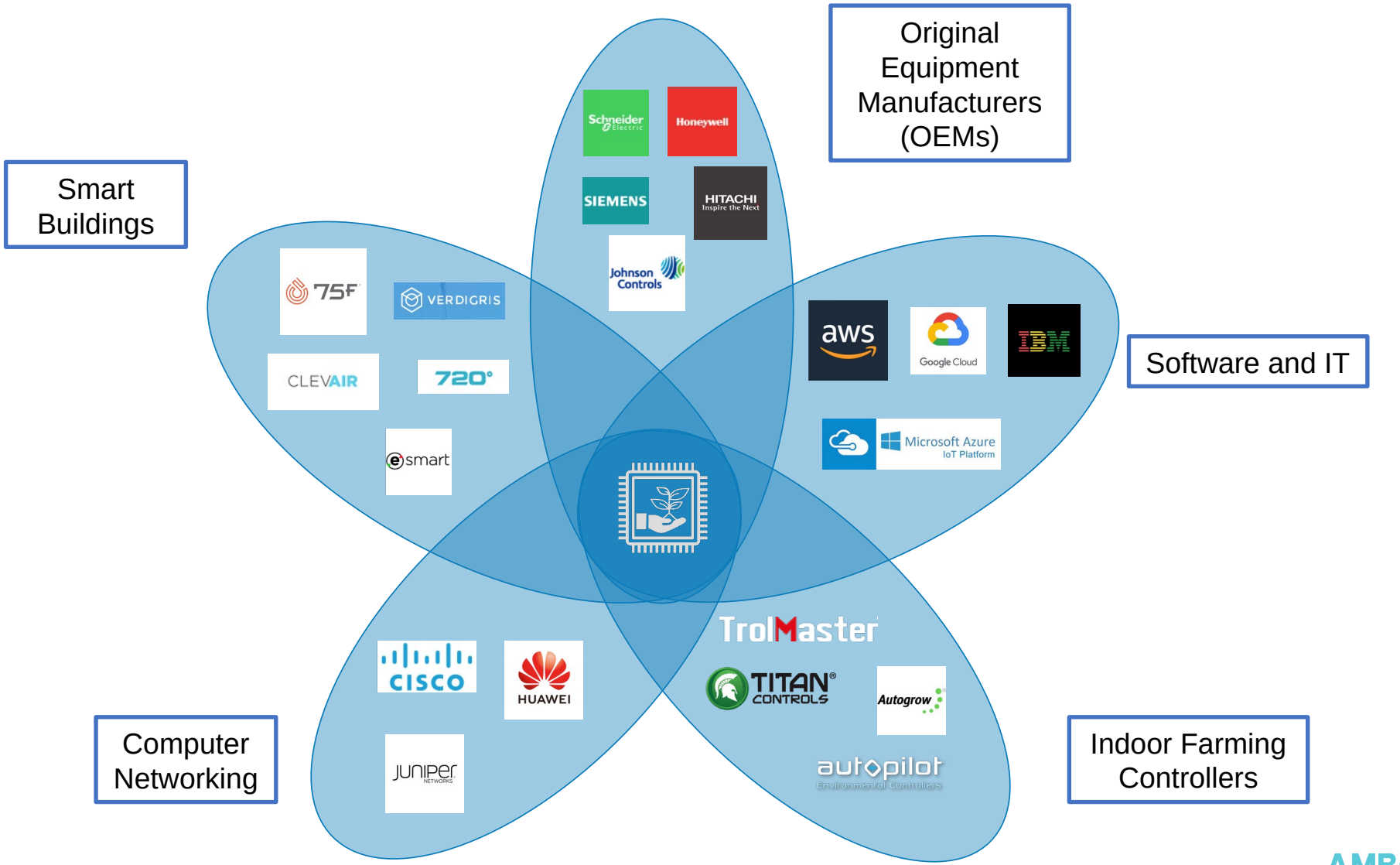
Ambient's Machine Learning Optimization Model

Patent



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Competitive Landscape: Turnkey Solution Provider

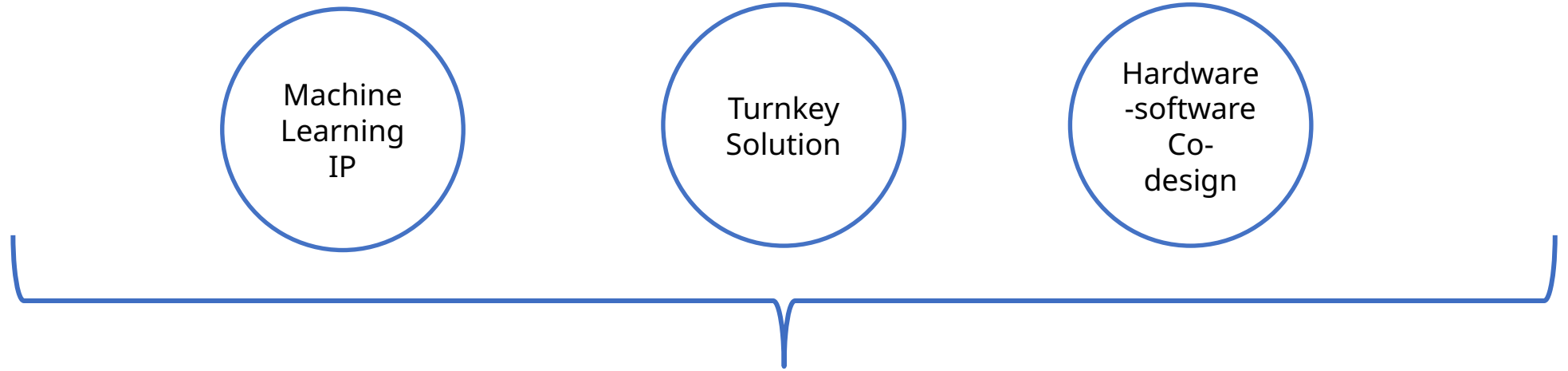


Ambient leverages hardware-software co-design to seamlessly connect Building and Indoor Farming systems

	AMBIENT SYSTEMS Climate At Your Service	Autogrow	75F	Johnson Controls
Segment	**Multiple**	**Indoor Farming Controller**	**Smart Buildings**	**OEM**
Indoor Farming Features	**Yes**	Yes	No	No
Modern Web-based UI	**Yes**	No	Yes	Some
Wireless IoT Network	**Yes**	No	Yes	Yes
Advanced Optimization Methods	**Machine Learning**	No	Some	Some

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Technology-enabled competitive advantage



Multiple Value Streams

Cost Reduction	Revenue Generation
Utility savings (energy efficiency)	Increase Yield
Reduction in manhours for highly trained engineers	Reduce Crop Loss (from sickness)
Lower insurance premiums (thanks to data access)	Enhanced Harvest Prediction

ness the power of data with our revenue sharing model

Pay If You Grow

ER UTILITY COST **INCREASED YIELD**

REDUCED MAN HOURS **HIGHER QUALITY**

Estimated Pricing Table for 1 Small Sized Farm
(220sqft or 20.4m2)

SGD($)

Implementation Cost	Hardware	**Quotation**
	Installation	**Quotation**
Operational Cost	Monthly Data Management	**Monthly**
	Monthly System Maintenance	**Monthly**
Use case specific	Monthly Estimated (% of Crop Value)	**5%**

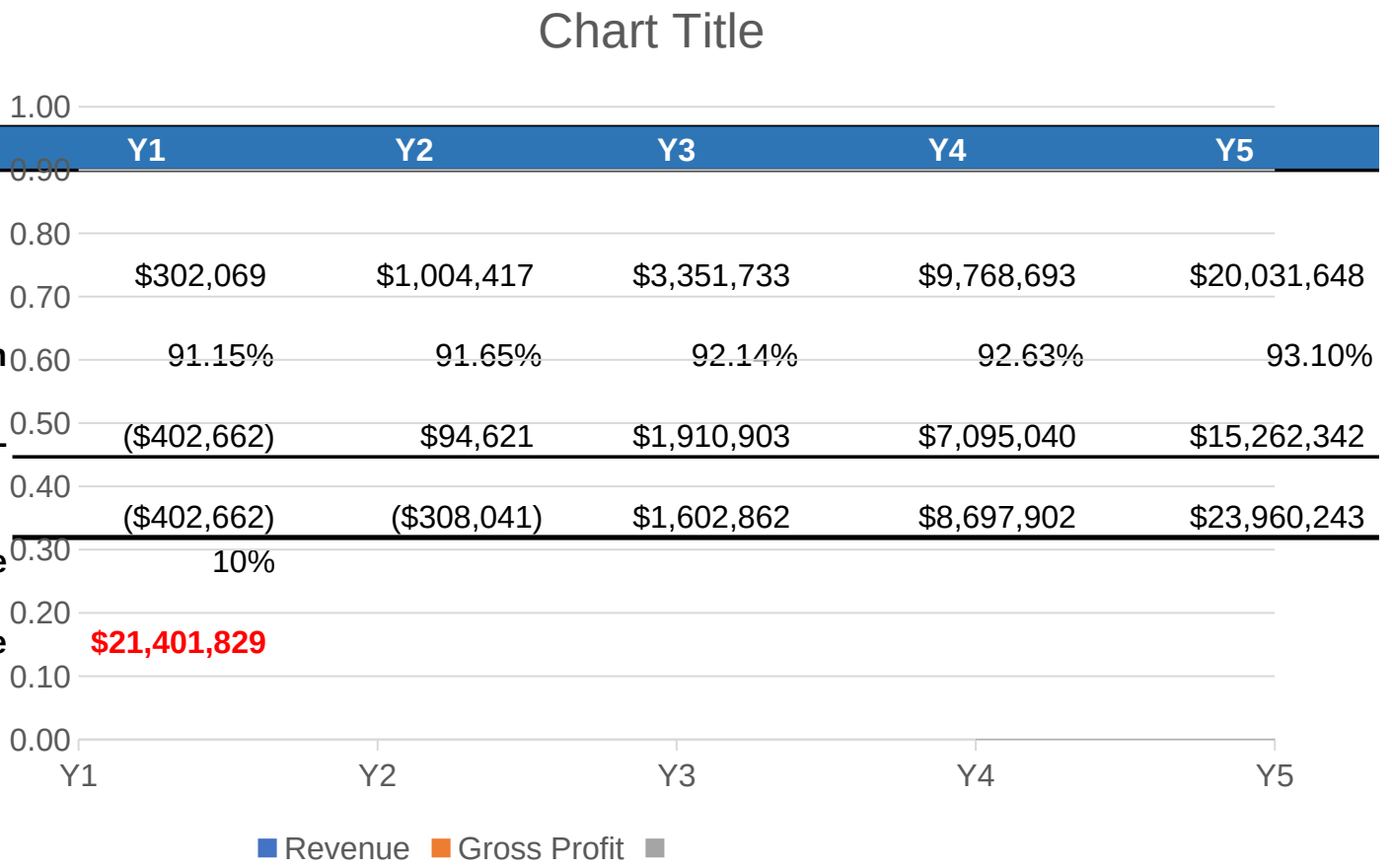


Private and Confidential

Our Financial Projections

[Adding Graph Here]

Base	Y1	Y2	Y3	Y4	Y5
Software Subscriptions (SGD$)					
Revenue	$302,069	$1,004,417	$3,351,733	$9,768,693	$20,031,648
Gross Margin	91.15%	91.65%	92.14%	92.63%	93.10%
P&L	($402,662)	$94,621	$1,910,903	$7,095,040	$15,262,342
	($402,662)	($308,041)	$1,602,862	$8,697,902	$23,960,243
Discount Rate	10%				
Net Present Value	**$21,401,829**				



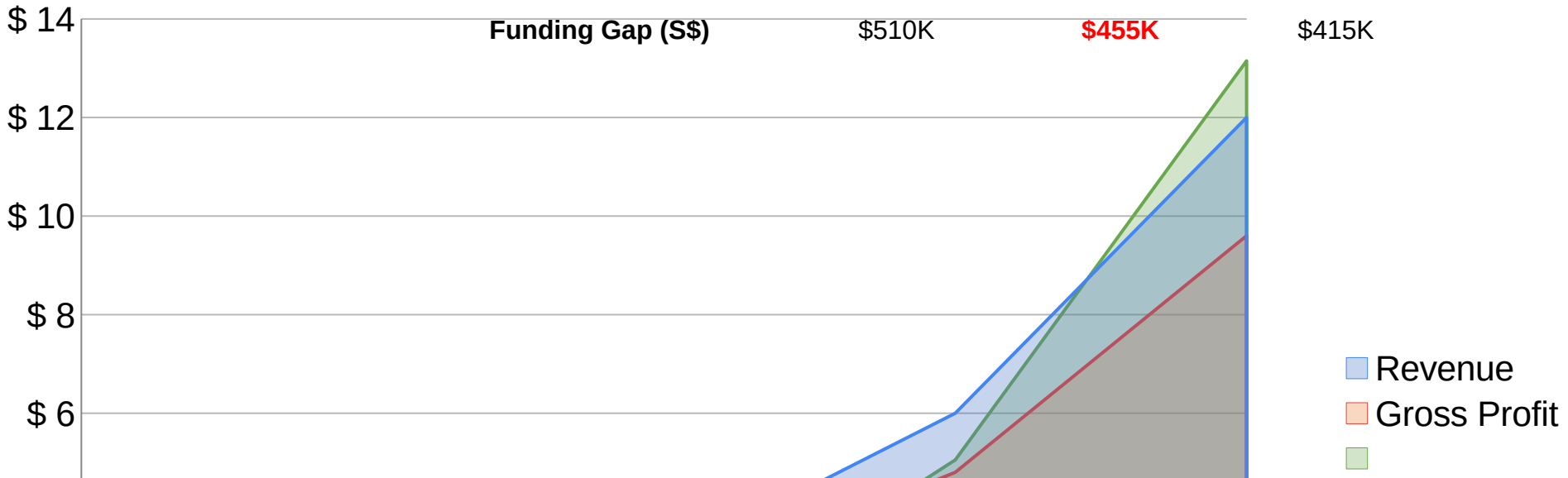
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Our Funding Ask and Use of Funds

[Adding Graph Here]

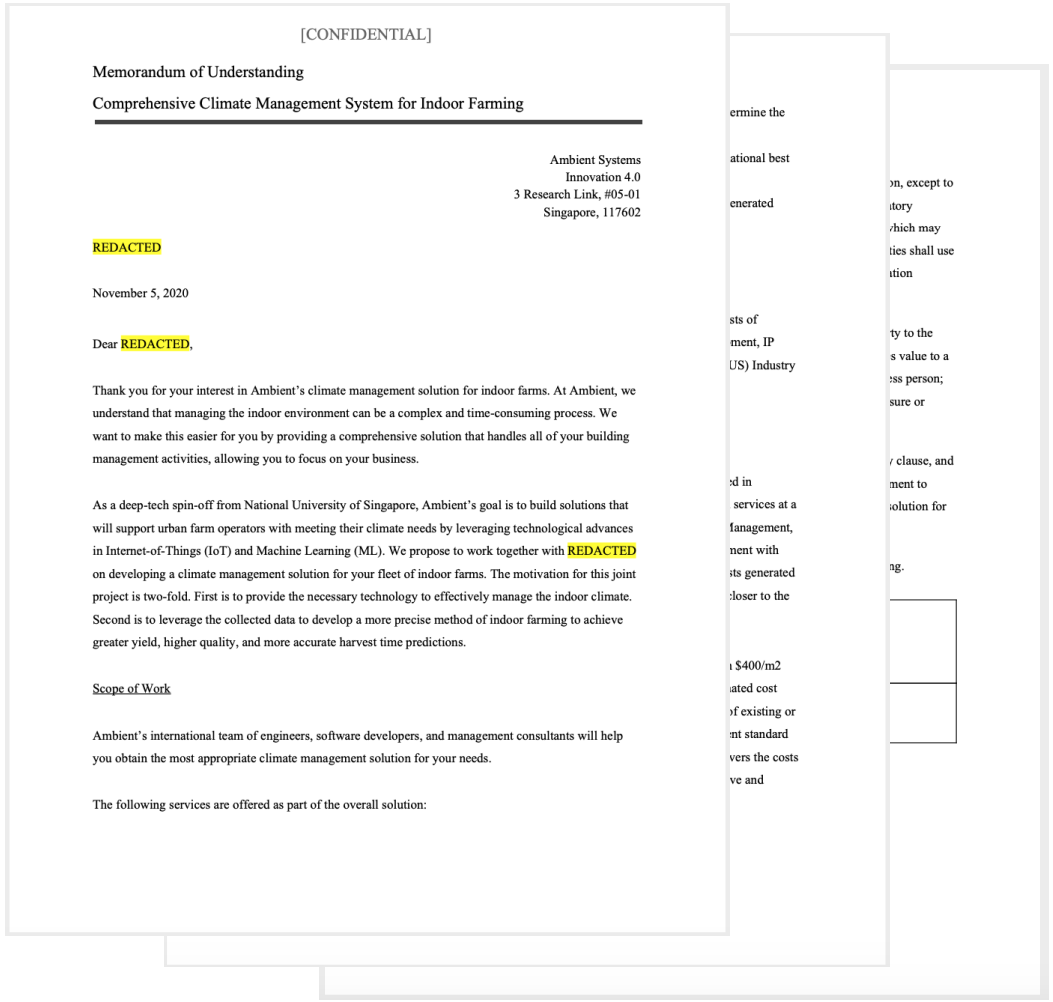
Base	Y1	Y2	Y3	Y4	Y5
Number of farms managed	14	45	141	354	566
Area of Farm Size Managed (m2)	1,164	3653	11,466	28,873	46,226
No. of Container Farms Equivalent	57	179	562	1415	2266

	Bear	Base	Bull
Runway (Months)	20	17	15
Funding Gap (S$)	$510K	$455K	$415K



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At Your Service

Customer Traction



[CONFIDENTIAL]

Memorandum of Understanding

Comprehensive Climate Management System for Indoor Farming

Ambient Systems
Innovation 4.0
3 Research Link, #05-01
Singapore, 117602

REDACTED

November 5, 2020

Dear REDACTED,

Thank you for your interest in Ambient's climate management solution for indoor farms. At Ambient, we understand that managing the indoor environment can be a complex and time-consuming process. We want to make this easier for you by providing a comprehensive solution that handles all of your building management activities, allowing you to focus on your business.

As a deep-tech spin-off from National University of Singapore, Ambient's goal is to build solutions that will support urban farm operators with meeting their climate needs by leveraging technological advances in Internet-of-Things (IoT) and Machine Learning (ML). We propose to work together with REDACTED on developing a climate management solution for your fleet of indoor farms. The motivation for this joint project is two-fold. First is to provide the necessary technology to effectively manage the indoor climate. Second is to leverage the collected data to develop a more precise method of indoor farming to achieve greater yield, higher quality, and more accurate harvest time predictions.

Scope of Work

Ambient's international team of engineers, software developers, and management consultants will help you obtain the most appropriate climate management solution for your needs.

The following services are offered as part of the overall solution:

1. **with a farm operator in**

2. **In conversation with two more leading farm operators**

3. **Received validation for a similar solution in the healthcare sector**

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Ambient's Climate As A Service solution can be seamlessly deployed in any industries that require a controlled environment









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Climate At Your Service

About Us

- **MComp, Computer Science**
  National University of Singapore
- **BASc, Civil Engineering, Sustainable Energy**
  University of Toronto
- **Serial Entrepreneur**
  5 years experience in cleantech & deep tech start-ups

, Finance
pore Management University

- **Strategy and Fundraising**
  3 years experience across e-commerce, gaming and payments



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1. Identified product-market fit as smart building technology provider for Indoor Farms

2. Competitive offering and IP through hardware-software co-design

3. Confirmed pilot project with farm operator

Appendix

The Future of Farming is Indoors



By 2050, the world is predicted to reach

9.7B

in population and to feed everyone would require increase in food output by

70%

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Traditional Farming Methods are increasingly unreliable



Global Warming has brought about issues such as

Rising Temperatures, Irregular Climates

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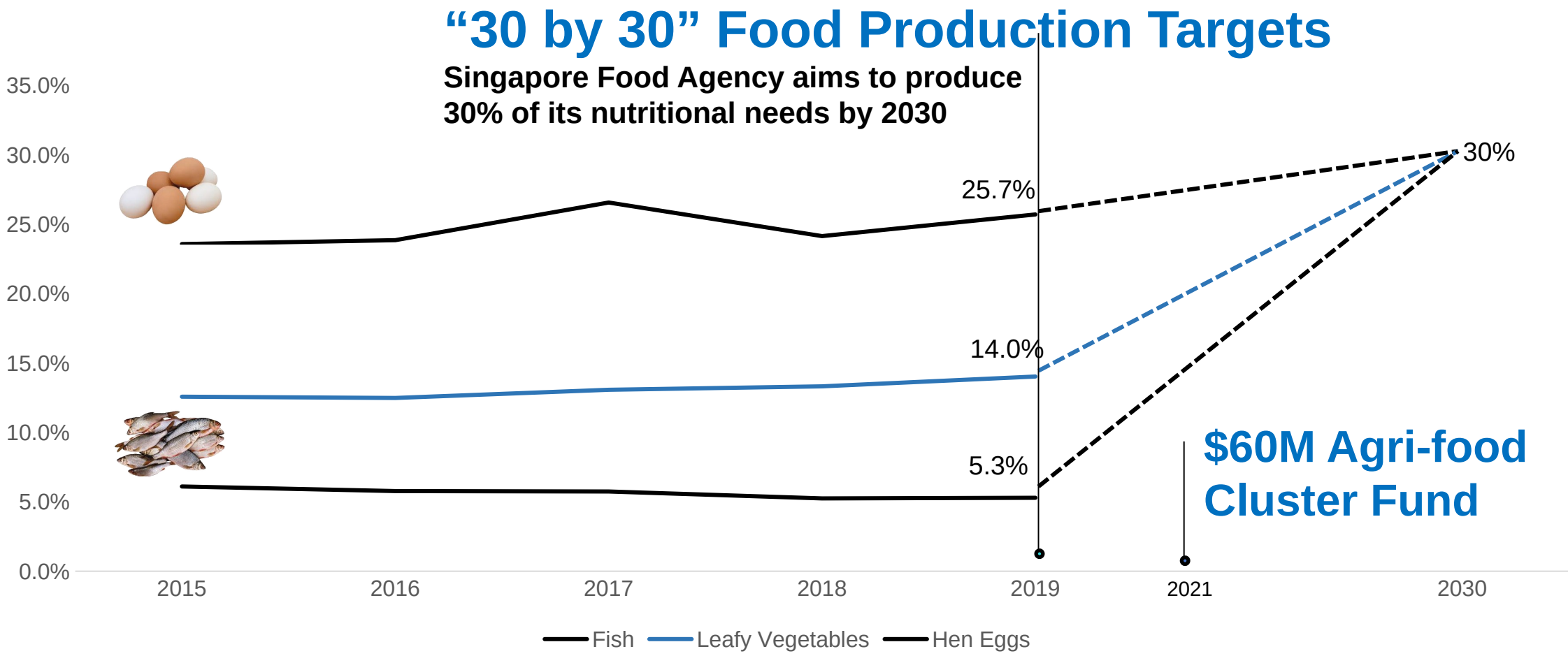
Covid-19 has disrupted traditional farming



Traditional Farms were hit hard by the Covid-19 Pandemic

Quarantines, Disruption to ,

Local farms sees supports from local government sectors



*Agriculture sector of Singapore is engaged mainly in the production of eggs, fish and vegetables for local consumptions
(1) Fish only includes seafood from coastal and land-based farm production and excluding local landings of seafood (seafood catches from local waters)
Source: Agri-food and Veterinary Authority, Singapore Food Agency, National Parks Board

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Our Business Model

Key Partners

NUS GRIP
Farm Operato
Building Owne
HVAC & BMS Ver
Government Agen

Key Activities

ements Gathering
rm Development
tem Installation
model design,
ment, training, and
tuning

Key Resources

IP
Platform
ML Model

Value Proposition

Initial: Enable successful climate management of vertical farms by measuring key parameters and ensuring process repeatability for future harvests.

data: increase yield, o
rgy use, and reduce la
costs.

Customer Relationships

Tailored Pricing
Modern Digital Platform
Personalized Support

Channels

Direct Outreach
Equipment Suppliers
Farmer Cooperatives

Customer Segments

Indoor Farms

Cost Structure

Revenue Streams

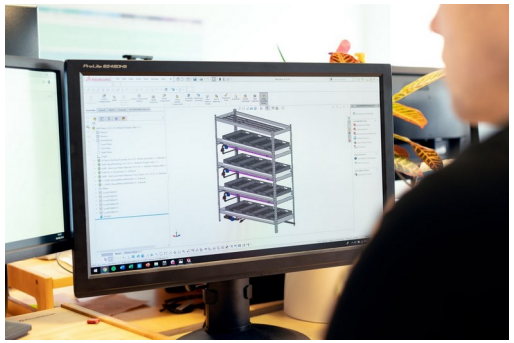
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Our Features



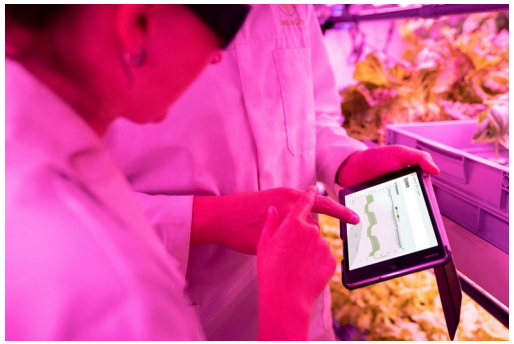
Seamless Building Integration

Enjoy full connectivity within your farm and perform real-time environmental adjustments based on crop



Monitoring In Real-Time

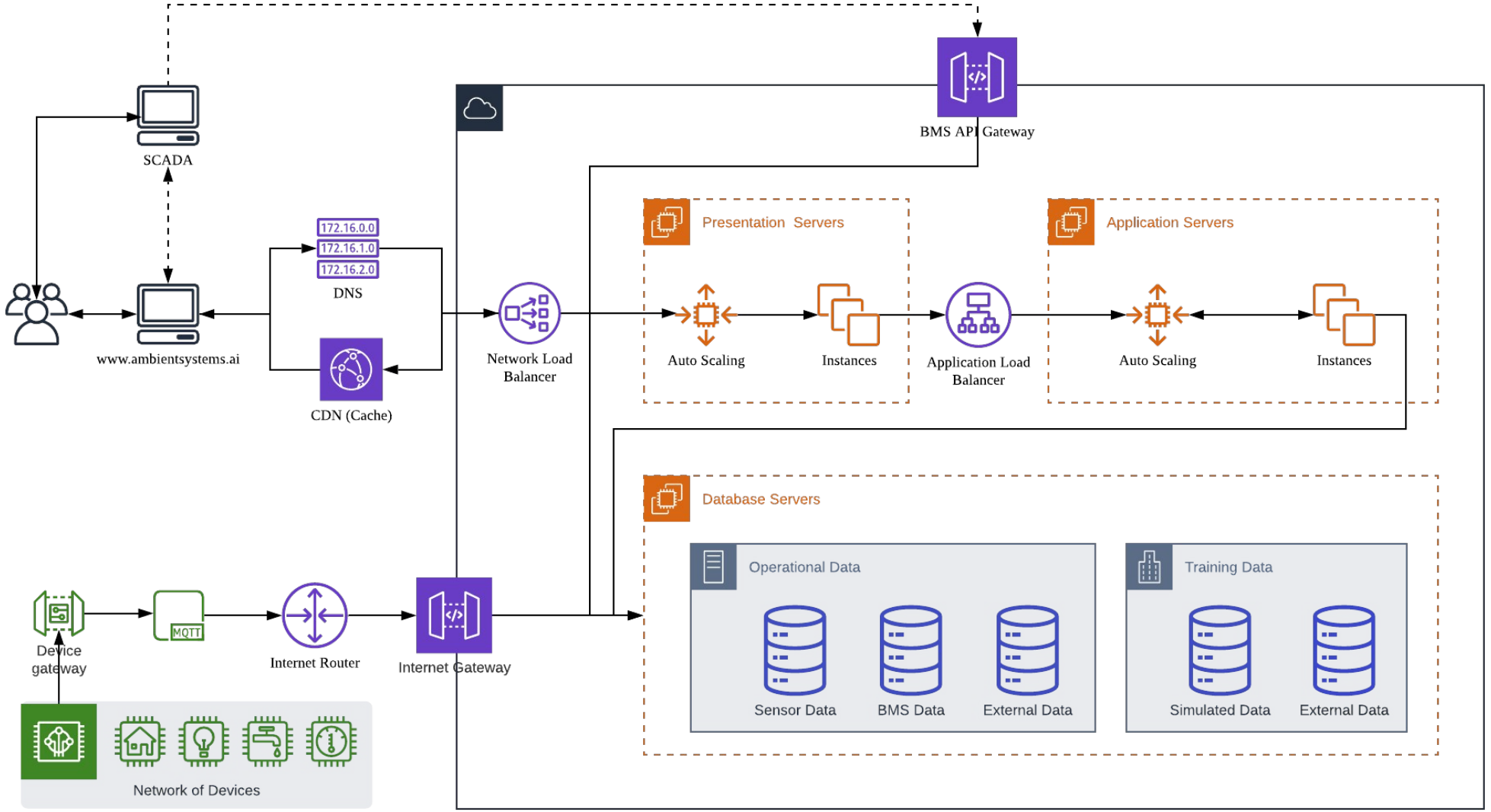
Access real-time data through online dashboards and receive push alerts anytime anywhere



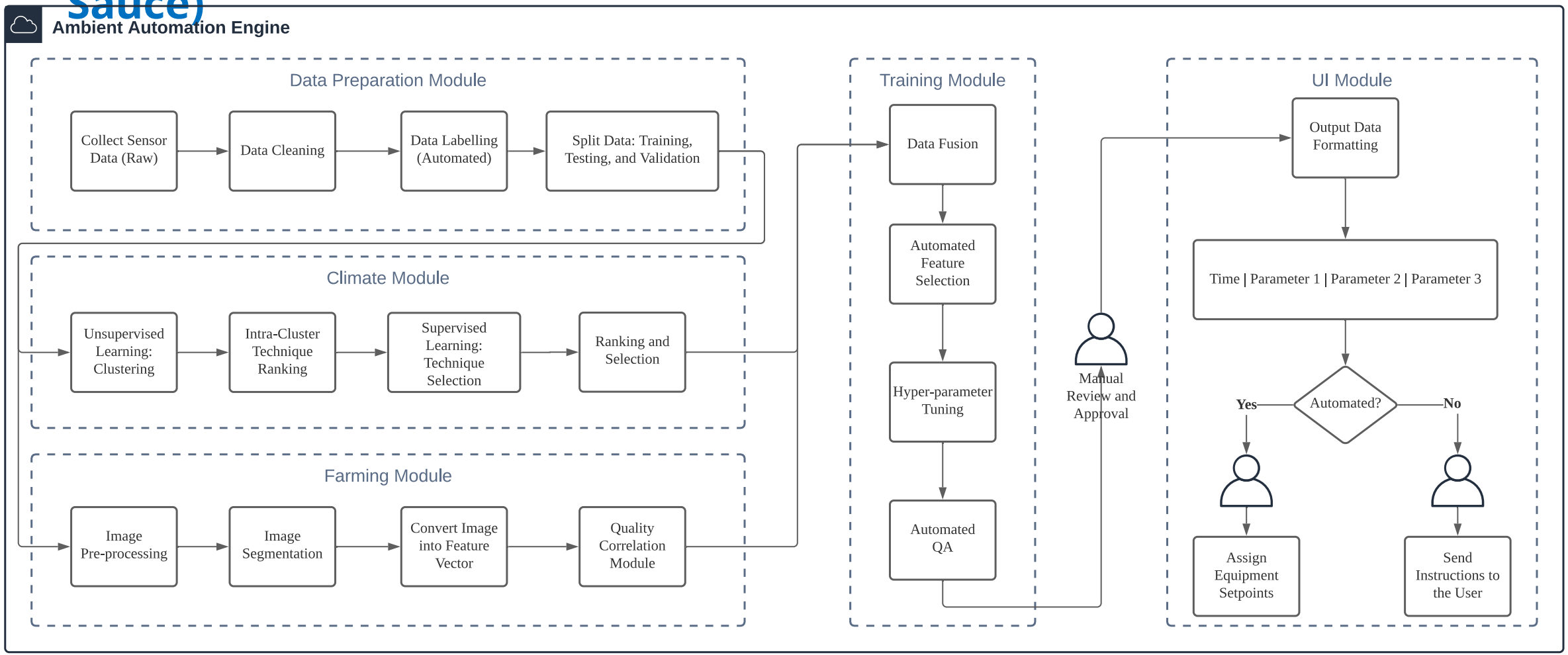
Grow With Data Insights

Apply data-driven solutions to understand factors that affect growth and quality, enabling better crop health

Our Technological Architecture



Deep dive into the Machine Learning Optimization Model (Secret Sauce)



Patent focuses on climate optimization specific to indoor farming

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Ambient aims to provide full-coverage of the Building Technology Ecosysten



2021

Software Subscription

2022

OEM Market



2023

2024

2025

Software Subscription: Data Visualization

Software Subscription: AI-enabled Optimization

Software Subscription: Controls/Automation

Sensor Assembly

Sensor Integration

Sensor Manufacturing

Controls Assembly

Controls Integration

Controls Manufacturing (OEM)

Building Technology



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